

APPLICATION DEVELOPMENT PROJECT REPORT
HOSPITAL MANAGEMENT SYSTEM

1. Student Details

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About Me :

I am a graduate in BSc Economics, Statistics and have just recently started my programming journey as a student of IITM Online BS . Having studied Mathematics and Statistics all along, I find pursuing a journey in the world of Data Science very interesting.

2. Project Details:

Title : Hospital Management System

Problem Statement :

To solve the problem of tracking and managing various variables in a hospital system including doctors, patients, appointments, bookings and treatments. The project aims to create a web application facilitating easy management and balancing of complex tasks.

Approach :

I created a REST Service for backend using Flask and SQLite3. The front end was implemented through a Flask app as well along with the usage of Bootstrap, CSS and Jinja2 for UI templates.

Swagger editor was used to manage APIs and to verify syntax of API spec.

The app allows booking of appointments, tracking doctor availability and uploading treatments along with other functionalities.

3. AI/LLM Declaration :

I used ChatGPT (GPT-5) for the following :

- Template generation and CSS : Generation of initial API spec : 5%
- Client side java script for form validation : 2%
- Synthetic data generation for testing app : 5%

The extent of AI/LLM usage is around 27 % in overall project development (split as above) limited to the above cases.

External help was taken for initial application design approach and review.

All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used :

Technology / Library	Purpose
Flask	Backend and frontend framework
SQLAlchemy	Connect flask and database to execute functions
Jinja2	HTML template rendering
Bootstrap 5	Presentation and visual aesthetics for UI
SQLite3	For backend database storage and management
Swagger Editor	For API Spec verification & backend testing

5. Database Scheme / ER Diagram:

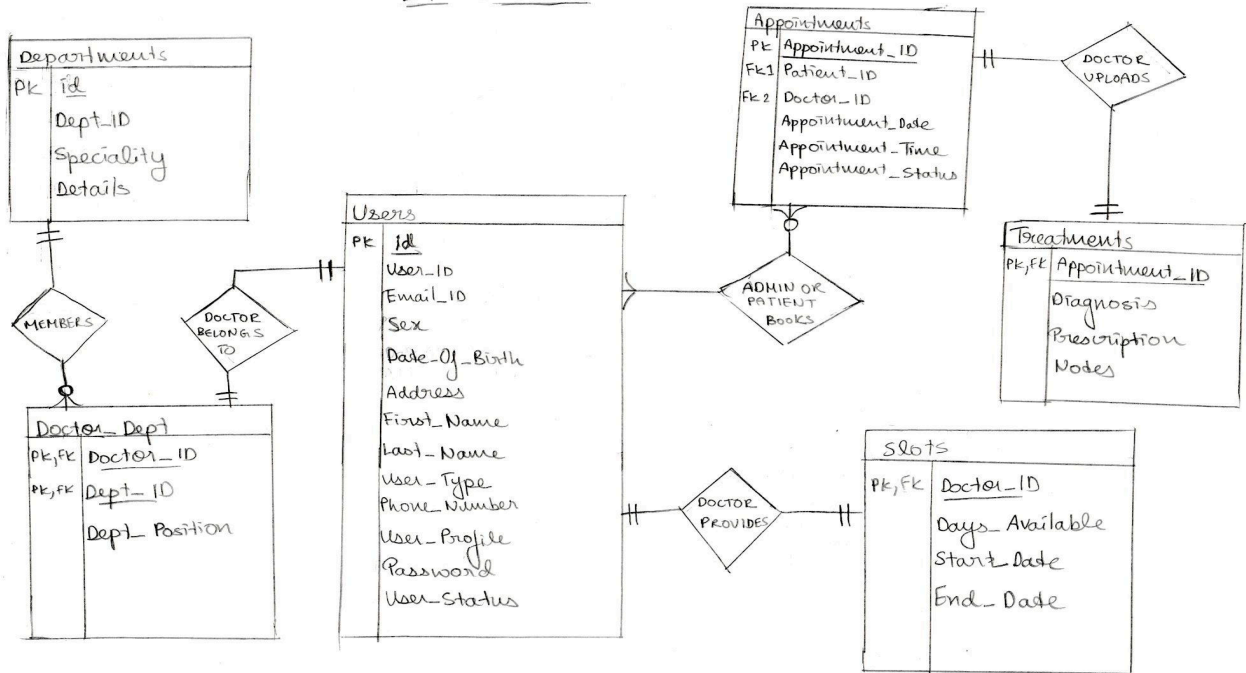
Tables:

- **Users** - Stores user details such as ID, Name, Age, DOB, Type, Status and contact details
- **Appointments** - Tracks all appointments scheduled, completed and cancelled and maintains record of doctor, patient, date and time.
- **Treatments** - Records treatment history and provides information relating to diagnosis, prescription and doctor notes for every appointment that is completed.
- **Departments** - Keeps track of all specialities and their details in the hospital.
- **Doctor_Dept** - Tracks the doctors belonging to each department
- **Slots** - Contains availability details for all doctors within a specific date range.

Relationships :

- Many- to - Many : Users -> Appointments
- One - to - One : Appointments -> Treatments
- One - to - One : Users -> Slots [User_Type : DOCTOR]
- One - to - One : Users -> Doctor_Dept [User_Type : DOCTOR]
- One - to Many : Departments -> Doctor_Dept [User_Type : DOCTOR]

HOSPITAL MANAGEMENT SYSTEM E-R DIAGRAM



6. API Resource Endpoints

Endpoint	Method	Description
/hms/id/generate/token	POST	Authenticate user and generate session token
/hms/user/create	POST	Create New User
/hms/user/lookup	GET	Lookup users by name or contact details
/hms/user/update	POST	update/edit user profiles
/hms/user/check-password	POST	Verify that password and hashed password match

/hms/user/delete	DELETE	Delete User from database
/hms/appointment/create	POST	Book appointment
/hms/appointments/lookup	GET	Lookup Appointments by date, status or Doctor/Patient ID
/hms/appointment/update	PUT	Make changes to appointment status
/hms/treatments/upload	POST	Upload Treatment
/hms/treatments/lookup	GET	Lookup treatments by patient id
/hms/treatments/update	POST	Update treatment record
/hms/departments/create	POST	Create new department
/hms/departments/lookup	GET	Lookup department by name or id
/hms/departments/delete	DELETE	Delete department
/hms/departments/assign	POST	Assign doctors to department
/hms/departments/unassign	DELETE	Remove doctors from department
/hms/departments/doctor-lookup	GET	Lookup doctors by speciality
/hms/slots/create	POST	Update availability for doctor
/hms/slots/lookup	GET	Lookup slots based on doctor id
/hms/slots/availability	GET	Returns free slots for a doctor based on date

OpenAPI Spec : [hms-openapi-spec.yaml](#)

7. **Architecture and Features:**

hms-backend/[app.py](#) - Backend Flask Application

hms-frontend/[app.py](#) - Front Flask Application

hms-frontend/templates - Html templates

sql - Synthetic data and DDL/Database Schema

hms-openapi-spec.yaml - Open API Spec

Implemented Features :

- User registration and login
- Appointment Booking
- View treatment history
- Department create for ADMIN
- Register doctor for ADMIN
- Secure password hashing and user session handling

8. **Video Presentation:**

Drive Link:

https://drive.google.com/file/d/1y2eNwkVBhFqgfnc_thzoKZhxiQu21oh/view?usp=drive_link